

Keenan Heinz

(925) 528-8095 | keenan6073@gmail.com | [linkedin.com/in/keenan-heinz](https://www.linkedin.com/in/keenan-heinz) | keenan6073.github.io/keenan-portfolio

EDUCATION

University of Pennsylvania, School of Engineering and Applied Sciences

Philadelphia, PA

Bachelor of Science in Engineering, Mechanical Engineering (Structures & Materials)

Aug. 2024 – May 2028

Relevant Coursework: Intro to Mechatronics, Solids, Dynamics, Thermodynamics I, Analytical Engineering Methods, Intro to Scientific Computing, Electromagnetics

ENGINEERING EXPERIENCE

Mechanical Engineering Intern

Menlo Park, CA

Stealth Startup

June 2026 – Present

- Owned mechanical, electrical, and software scope for a fleet of automated test rigs, from design criteria through fabrication-ready documentation.

Automated Test Rig Fleet (Mini-Instron DAQ Systems)

- Authored 500+ line design spreadsheets covering design criteria, BOMs, and hand calculations.
- Ran fastener and structural calculations to a 2.0 yield safety factor across all fixture designs, rated to a 50 kg maximum working load.
- Built wiring and control stacks: STM32 motion control, precision ADC, and load-cell signal chains.
- Wrote SPI-based DAQ software for synchronized force, displacement, and time data capture across the fleet.
- Self-taught design for manufacturing (DFM) practices to prepare designs for outsourced fabrication.

Non-Contact Optical Extensometer

- Selected lens and working-distance optics for non-contact strain measurement systems.
- Used AI-assisted tools to develop vision-processing software and camera-controller synchronization for accurate post-processing.

Coil Winding System (ITC-PLC-70)

- Designing custom bobbins and a tensioning system for consistent wire feed across varying coil geometries.
- Building HMI-integrated software to set and adjust coil requirement parameters on the fly.
- Running test winds to validate machine scalability for future production volume.

Co-Lead Engineer

Philadelphia, PA

Penn Hyperloop

Sep. 2025 – Present

- Designed an A36 steel propulsion structure for an in-house Tunnel Boring Machine, in Fusion 360, rated for vertical loads >2500 N and horizontal loads >5000 N.
- Ran FEA and analytical hand calculations to validate structural integrity under worst-case loading.
- Evaluated 50+ suppliers for actuator sourcing as part of a full system redesign, achieving 150 kN thrust and 1+ meter stroke targets while minimizing procurement cost.
- Promoted from Propulsion Engineer to Co-Lead Engineer after two academic semesters; now directs mechanical and electrical integration across sub-teams for the 2027 competition machine.

ATHLETICS

D1 and National Team Athlete, Rowing

Philadelphia, PA

University of Pennsylvania / USRowing National Team

Aug. 2024 – Present

- Two-time National Championship qualifier: 8th, 2026 IRA; 9th, 2025 IRA (team best points finish).
- 2nd, 2024 World Rowing Championships (St. Catharine's); 5th, 2023 World Rowing Championships (Paris).

TECHNICAL SKILLS

CAD & Simulation: SolidWorks, Fusion 360, Onshape, Finite Element Analysis, DraftSight, MATLAB

Electrical & Controls: wiring, signal chains, STM32, stepper drivers, ADCs, load cells (AI-assisted)

Manufacturing: DFM, production engineering (self-taught), MIG welding, ironworker, band saws, drill press

Programming: Python, Java, Excel, spreadsheet-based engineering calculations

Design: DaVinci Resolve, CapCut, Adobe Photoshop, Pixlr, PowerPoint, Canva

Interests: backpacking and camping, road biking, classic rally homologation cars, rock music, cats